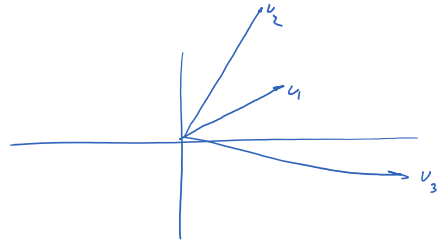
Some Definitions

Def Linear independence

A set of vectors $v_1, v_2, \dots, v_n$ is linearly independent if the linear combination

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

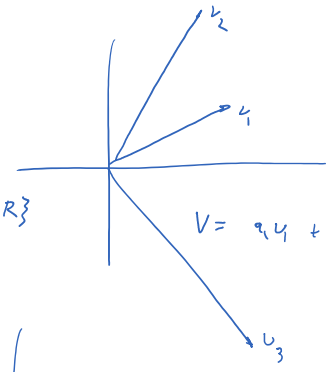
only when $a_1 = a_2 = \dots = a_n = 0$.



$v_3 = a_2 v_2 + a_1 v_1$

Def: span

$$\text{span}((v_1, v_2 \dots v_n)) = \{ v \in S : v = a_1 v_1 + \dots + a_n v_n \text{ for some } a_1, a_2, \dots, a_n \in \mathbb{R} \}$$

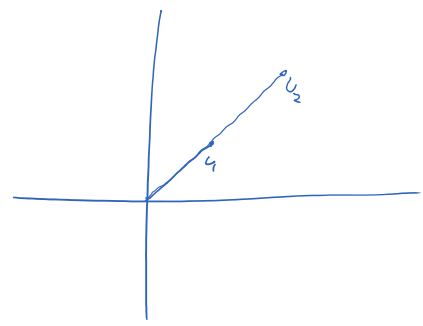


$V = a_1 v_1 + a_2 v_2$

Def: spanning set.

we say that a set $T = \{ v_1, v_2, \dots, v_n \}$ is a spanning set for a vector space V if $V = \text{span}(T)$.

Def: A basis of a v.s. V is a spanning set that is linearly independent.



Def: The dimension of a vectors space V is the number of elements of a basis for V.

If T is a basis for V : $\dim(V) = |T|$

Notation: Vectors $\begin{matrix} | & | & & | \\ p_1, & p_2, & \dots, & p_n \\ | & | & & | \end{matrix}$ where each $p_i \in \mathbb{R}^m$

Form $A = \begin{bmatrix} | & | & & | \\ p_1 & p_2 & \cdots & p_n \\ | & | & & | \end{bmatrix}_{m \times n}$

linear combination

$$\underline{x} = c_1 p_1 + c_2 p_2 + \dots + c_m p_m$$

$$= \underbrace{[\, p_1 \ p_2 \ \cdots \ p_n \,]}_{A} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$= \quad A \subseteq \qquad \subseteq$

Def: Let V be a vector space. Then a real-valued function

$$\|\cdot\| : V \longrightarrow \mathbb{R}$$

is a <u>norm</u> if it satisfies the following:

N1: $\|x\| \geq 0 \quad \forall\, x \in V$

N2: $\|x\| = 0$ iff $x = 0$

N3: $\|\alpha x\| = |\alpha| \cdot \|x\|, \quad \forall\, \alpha \in \mathbb{R},\ x \in V$

N4: $\|x+y\| \leq \|x\| + \|y\|$ (triangle inequality)

For a metric space that is a v.s.

$$d(x,y) = \|x-y\|$$

$$\|x\| = d(x, 0)$$

In $\mathbb{R}^d$:

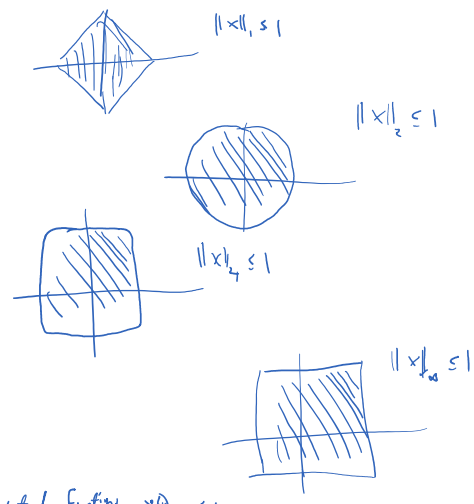
<u>Ex:</u>

$$\|x\|_1 = \sum_{i=1}^{d} |x_i| \qquad \ell_1$$

$$\|x\|_2 = \left(\sum_{i=1}^{d} |x_i|^2 \right)^{1/2} \qquad \ell_2$$

$$\|x\|_p = \left[\sum_{i=1}^{d} |x_i|^p \right]^{1/p} \qquad \ell_p$$

$$\|x\|_\infty = \max_{1 \leq i \leq d} |x_i| \qquad \ell_\infty$$



Ex: V = a space of functions.

$$\|x(t)\|_1 = \int_a^b |x(t)|\, dt$$

L_1: the set of functions $x(t)$ s.t. $\|x(t)\|_1 < \infty$

$$\|x(t)\|_2 = \left(\int_a^b |x(t)|^2\, dt \right)^{1/2}$$

L_2: the set of functions $x(t)$ s.t. $\|x(t)\|_2 < \infty$

$L_2[a,b]$

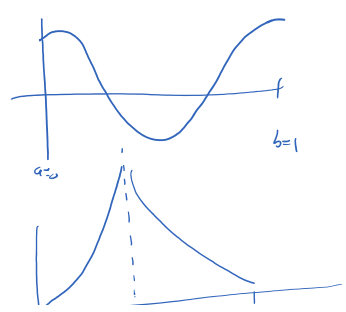
$L_2(-\infty, \infty)$.

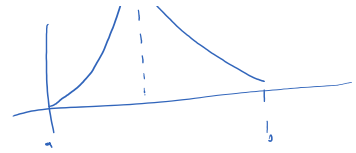
$$\|x(t)\|_p = \left(\int_a^b |x(t)|^p\, dt \right)^{1/p}$$

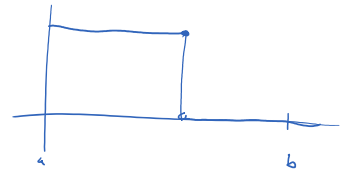
$$\|x(t)\|_\infty = \sup_{t \in [a,b]} |x(t)|.$$

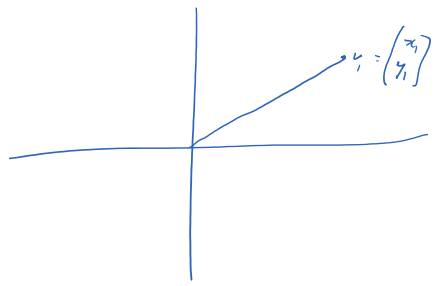
Q: $C[a,b]$

$L_2(a,b)$









$\| v_1 \|_2 = \sqrt{x_1^2 + y_1^2}$

Inner Product spaces

two slots

Def: Let V be a v.s. A function $\langle \, , \, \rangle : V \times V \longrightarrow \mathbb{R}$ is an <u>inner product</u> if it satisfies the following:

IP1: $\langle x, y \rangle = \langle y, x \rangle$ (real) $\qquad \langle x, y \rangle = \langle y, x \rangle^*$ (complex)

IP2: $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ $\qquad \langle x, \alpha y \rangle = \alpha \langle x, y \rangle$ (real)

α : scalar. $\qquad \langle x, \alpha y \rangle = \alpha^* \langle x, y \rangle$ (complex)

IP3: $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

IP4: $\langle x, x \rangle > 0$ if $x \neq 0$

$\langle x, x \rangle = 0$ iff $x = 0$

Ex: $\underline{x}, \underline{y} \in \mathbb{R}^n$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad \underline{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\langle \underline{x}, \underline{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$
$$= \sum_{i=1}^{n} x_i y_i$$
$$= \underline{x}^T \underline{y} = \underline{y}^T \underline{x}.$$

Ex: $\underline{x}, \underline{y} \in \mathbb{C}^n$

$$\langle \underline{x}, \underline{y} \rangle = \sum_{i=1}^{n} x_i y_i^*$$
$$= \underline{y}^H \underline{x}.$$

Ex: $L_2[0,1]$ Real

$x(t), y(t) \in L_2[0,1]$

$$\langle x(t), y(t)\rangle = \int_0^1 x(t)y(t)\,dt$$

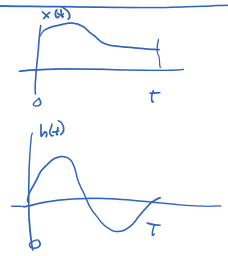
$L_2[0,1]$ complex

$$\langle x(t), y(t)\rangle = \int_0^1 x(t)\, y^*(t)\,dt$$

Ex: $L_2(-\infty, \infty)$

$$\langle x(t), e^{j\omega t}\rangle = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\,dt$$

Ex:



$$y(T) = x(t) * h(t)\Big|_{t=T}$$

$$= \int_0^T x(\tau)\, h(T-\tau)\,d\tau$$

Introduce: $g(\tau) = h(T-\tau)$

$$y(T) = \int_0^T x(\tau)\, g(\tau)\,d\tau$$

$$= \langle x(t), g(t)\rangle$$

Ex: $A \in \mathbb{R}^{m\times n}$

$B \in \mathbb{R}^{m\times n}$

$$\langle A, B\rangle = tr(B^T A)$$

$A \in \mathbb{C}^{m\times n}$

$B \in \mathbb{C}^{m\times n}$

$$\langle A, B\rangle = tr(B^H A).$$

Induced Norms

$$\|x\|_2 = [\langle x, x\rangle]^{1/2}$$

Ex: Let $R_{d\times d}$ be a positive definite matrix, p.d.

$R > 0$

(i.e., $\underline{x}^T R \underline{x} > 0$ for all $\underline{x} \neq 0$)

Also, Let R be symmetric.

Define $\langle\, ,\, \rangle : \mathbb{R}^d \times \mathbb{R}^d \longrightarrow \mathbb{R}$

by $\underline{\langle \underline{x}, \underline{y}\rangle = \underline{y}^T R \underline{x}}$

$$\|x\|_2^2 = \langle x, x\rangle$$

$$= \underline{x}^T R \underline{x}$$

$$(\underline{y} - A\underline{x})^T Q^{-1} (\underline{y} - A\underline{x})$$

$$= \langle y - Ax, y - Ax\rangle, \qquad R = Q^{-1}$$

$$\|y - Ax\|_2^2$$

by $\underbrace{\langle \underline{x}, \underline{y} \rangle = \underline{y}' R \underline{x}}$

$$\|x\|_2^2 = \langle x, x \rangle$$
$$= \underline{x}^T R \underline{x}$$

$$(\underline{y} - A\underline{x})^T Q^{-1} (\underline{y} - A\underline{x})$$
$$= \langle y - Ax, y - Ax \rangle, \qquad R = Q^{-1}$$
$$\|y - Ax\|_2^2$$

Ex: $L_2[-1, 1]$ $\qquad \langle x(t), y(t) \rangle = \int_{-1}^{1} x(t) y(t)\, dt$

Let $w(t)$ be a function,

$w(t) \geq 0 \quad \forall t \in [-1, 1]$

except at possibly a countable number of points.

$$\langle x(t), y(t) \rangle = \int_{-1}^{1} w(t) x(t) y(t)\, dt$$

$$\|x(t)\|_2^2 = \langle x(t), x(t) \rangle$$

Cauchy-Schwartz Inequality

$$\left[\int_a^b x(t) y(t)\, dt \right]^2 \leq \int_a^b x(t)^2\, dt \int_a^b y(t)^2\, dt$$

$$\cancel{\left[\int_a^b x(t) y(t)\, dt \right]^2 = \int_a^b x(t)^2 y(t)^2\, dt}$$

Thm: $\boxed{|\langle x, y \rangle| \leq \|x\| \cdot \|y\| \quad \text{where } \|\cdot\| \text{ is induced from } \langle \cdot, \cdot \rangle}$

with equality iff $y = \alpha x$ for some scalar α.

$$\left| \sum_{i=1}^{n} x_i y_i \right| \leq \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2} \left(\sum_{i=1}^{n} |y_i|^2 \right)^{1/2}$$

Pf: Consider $\|x - \alpha y\|^2$. $\qquad$ we know $\|x - \alpha y\|^2 \geq 0$

$$\|x - \alpha y\|^2 = \langle x - \alpha y, x - \alpha y \rangle$$

$$= \langle x, x\rangle - \langle x, \alpha y\rangle - \langle \alpha y, x\rangle + \langle \alpha y, \alpha y\rangle$$

$$= \|x\|^2 - \alpha^* \langle x, y\rangle - \alpha \langle y, x\rangle + |\alpha|^2 \langle y, y\rangle$$

$$= \|x\|^2 - 2\,\mathrm{Re}\left[\alpha^* \langle x, y\rangle\right] + |\alpha|^2 \|y\|^2$$

(complete the square)

$$= \|y\|^2 \left(\alpha - \frac{\langle x,y\rangle}{\|y\|^2}\right)\left(\alpha^* - \frac{\langle y,x\rangle}{\|y\|^2}\right) + \|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2}$$

$$\geq 0$$

choose the α to minimize this, $\alpha = \frac{\langle x,y\rangle}{\|y\|^2}$

$$\|x\|^2 - \frac{|\langle x,y\rangle|^2}{\|y\|^2} \geq 0$$

$$\boxed{\|x\|^2 \|y\|^2 \geq |\langle x,y\rangle|^2}$$

Now set $y = \alpha x$

$$|\langle x,y\rangle| = |\langle x, \alpha x\rangle|$$

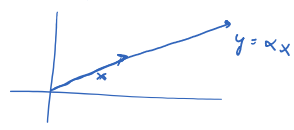
$$= |\alpha| \langle x, x\rangle = |\alpha| \|x\|^2$$

$$\|x\|^2 \cdot \|y\|^2 = \|x\|^2 \cdot \|\alpha x\|^2 = \|x\|^2 \cdot |\alpha|^2 \cdot \|x\|^2$$

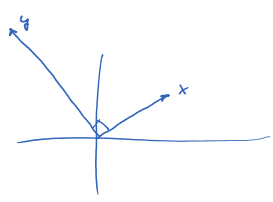
Think about $\boxed{-1 \leq \frac{\langle x,y\rangle}{\|x\| \cdot \|y\|} \leq 1}$

Define an angle θ s.t. $\boxed{\cos\theta = \frac{\langle x,y\rangle}{\|x\| \cdot \|y\|}}$.

<u>Def:</u> If $x = \alpha y$, x and y are said to be <u>colinear</u>. In this case, $\cos\theta = 1$



<u>Def.</u> If $\langle x,y\rangle = 0$, x and y are said to be orthogonal. $\perp$



$\cos\theta = 0$

Def: If $\underline{v}_1, \underline{v}_2, \ldots, \underline{v}_m$, and $\langle \underline{v}_i, \underline{v}_j\rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

then set is said to

<u>orthonormal</u>

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ex: $[-1, 1]$ $\quad x(t) = 1 + t^2$

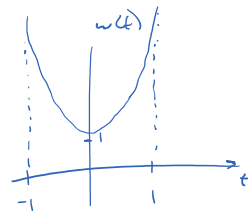
ex: $L_2[-1,1]$ $\quad x_1(t) = 1 + t^2$

$x_2(t) = t^2 - 2t$

$$\cos\theta = \frac{\langle x_1(t), x_2(t)\rangle}{\|x_1(t)\| \cdot \|x_2(t)\|} = \frac{\int_{-1}^{1} x_1(t)\, x_2(t)\, dt}{\left(\int_{-1}^{1} x_1(t)^2\, dt\right)^{1/2} \left(\int_{-1}^{1} x_2(t)^2\, dt\right)^{1/2}}$$

Ex: $L_2[-1,1]$ $\quad \langle x(t), y(t)\rangle = \int_{-1}^{1} w(t)\, x(t)\, y(t)\, dt$

where $w(t) = \dfrac{1}{\sqrt{1-t^2}}$

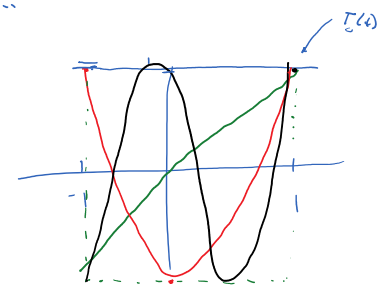


Let $T_n(t) = \cos(n \cos^{-1}(t))$, $\quad t \in [-1,1]$

$n = 0, 1, 2, \ldots$

$n=0$: $\quad T_0(t) = \cos(0) = 1$

$n=1$ $\quad T_1(t) = \cos(\cos^{-1}(t)) = t$

$n=2$ $\quad T_2(t) = \cos(2\cos^{-1}(t)) = 2t^2 - 1$

$n=3$ $\quad T_3(t) = \cos(3\cos^{-1}(t)) = 4t^3 - 3t$